# Introduction to Clustering

Lieven Clement

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# 1 Introduction

## 1.1 Objective

Objective: grouping of observations into **clusters**, so that

- similar observations appear in the same cluster
- dissimilar observations appear in distinct clusters

$\longrightarrow$ need for a measure for **similarity** and **dissimilarity**?

### 1.2 Example 1

Single cell transcriptomics: $n \times p$ Matrix for which

- every column contains the expression levels of one of $p$ genes for $n$ cells
- every row contains the expression levels of $p$ genes for one cell (**sample**)
- Research question: look for groups of cells that have similar gene expression patterns
- Or, look for groups of genes that have similar expression levels across the different cells. This can help us in understanding the regulation and functionality of the genes.

$\longrightarrow$ both **observations** (rows) and **variables** (columns) can be clustered

### 1.3 Example 2.

Abundance studies: the abundances of $n$ plant species are counted on $p$ plots (habitats)

- look for groups that contain species that live in the same habitats, or, look for groups of habitats that have similar species communities

$\longrightarrow$ both **observations** (rows) and **variables** (columns) can be clustered

## 2 Partition Based Cluster Analysis

- Partition based cluster methods require the number of clusters (k) to be specified prior to the start of the algorithm.

### 2.1 K-means Methods

- To use the k-means clustering algorithm we have to pre-define $k$, the number of clusters we want to define.
- The k-means algorithm is iterative.
- The algorithm starts by defining k cluster centers (centroids).
- Then the algorithm proceeds as follows

  1. First each observation is assigned to the cluster with the closest center to that observation.
  2. Then the k centers are redefined using the observations in each cluster, i.e. the multivariate means (column means) of all observations in a cluster are used to define each new cluster center.
  3. We repeat these two steps until the centers converge.

## 2.2 Example

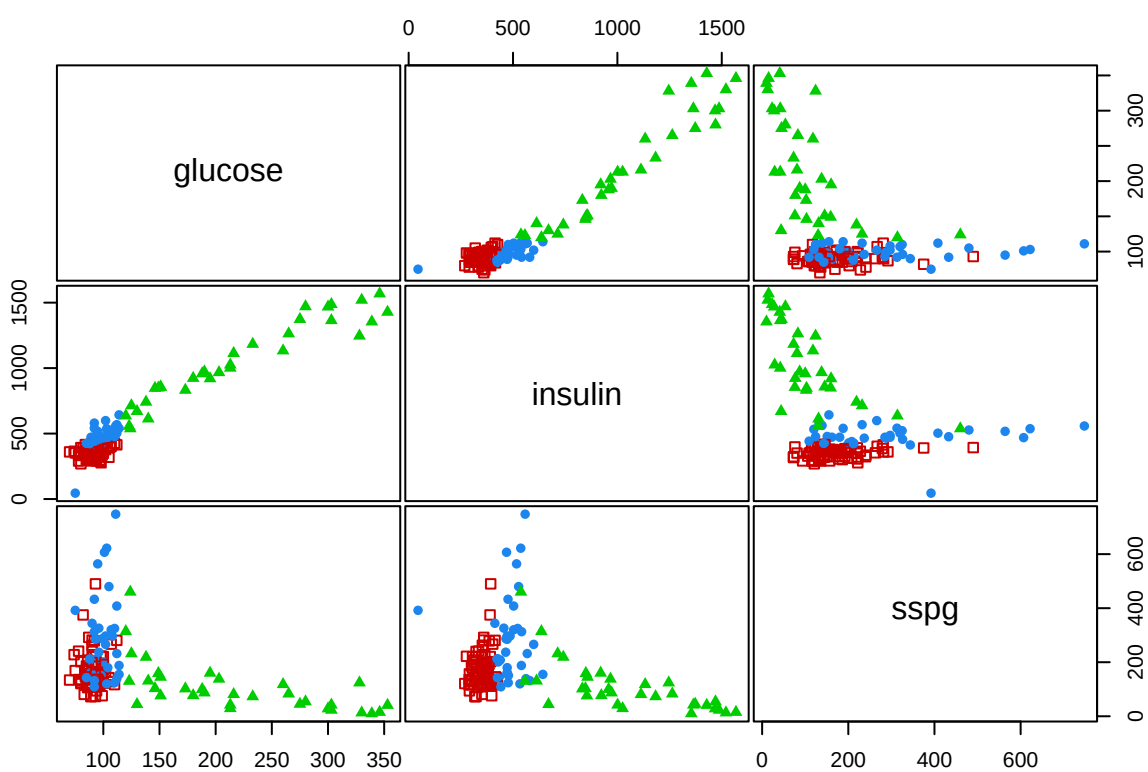
```r
library(tidyverse)
data(diabetes,package = "mclust")
class <- diabetes$class
table(class)
```

```
## class
## Chemical   Normal    Overt
##       36       76       33
```

```r
head(diabetes)
```

```
##    class glucose insulin sspg
## 1 Normal      80     356  124
## 2 Normal      97     289  117
## 3 Normal     105     319  143
## 4 Normal      90     356  199
## 5 Normal      90     323  240
## 6 Normal      86     381  157
```

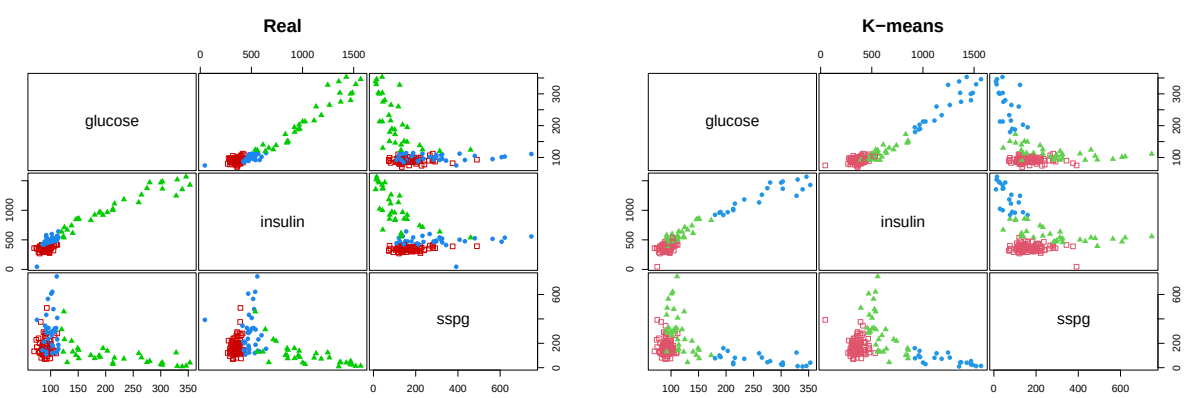
```r
mclust::clPairs(diabetes[,-1], diabetes$class)
```



```r
diabetesKmeans <- kmeans(diabetes[,-1], centers = 3)
diabetesKmeans
```

```
## K-means clustering with 3 clusters of sizes 90, 34, 21
##
## Cluster means:
##     glucose   insulin      sspg
## 1  91.55556  362.0000 173.05556
## 2 115.73529  591.1176 294.97059
## 3 262.52381 1225.5238  65.85714
##
## Clustering vector:
##   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
##  21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
##  41  42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
##  61  62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
##  81  82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99 100
##   1   2   1   1   1   2   2   2   2   2   2   2   2   2   2   1   2   2   2   2
## 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120
##   2   2   1   1   1   2   2   2   2   1   2   1   3   3   2   3   3   3   2   3
## 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140
##   2   3   3   2   2   3   3   2   3   2   2   3   3   2   2   2   2   3   3   3
## 141 142 143 144 145
##   3   3   3   3   3
##
## Within cluster sum of squares by cluster:
## [1]  699088.9 1571557.1 1098989.0
##  (between_SS / total_SS =  80.6 %)
##
## Available components:
##
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

```r
mclust::clPairs(diabetes[,-1], diabetes$class, main = "Real"); mclust::clPairs(diabetes[,-1], diabetesKm
```



# 3 Hierarchical Cluster Analysis

## 3.1 General Algorithm

- In step 0 each observations is considered as a cluster (i.e. $n$ clusters).
- Every next step consists of:

  1. merge the two clusters with the smallest intercluster dissimilarity
  2. recalculate the intercluster dissimilarities

In step 0 the intercluster dissimilarity coincides with the dissimilarity between the corresponding observations

$\rightarrow$ intercluster dissimilarity?

## 3.2 Intercluster Dissimilarities

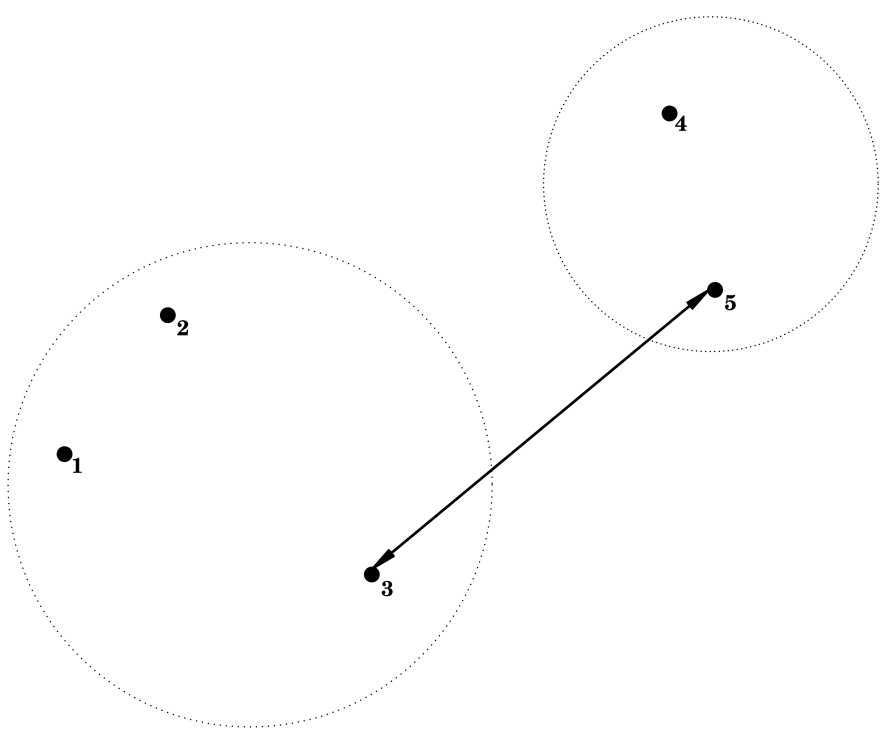
- Represent clusters (e.g. $C_1$ and $C_2$) as sets of points $\mathbf{x}_i$ which belong to that cluster
- $d(C_1, C_2)$: intercluster dissimilarity between

We consider three intercluster dissimilarities.

### 3.2.1 Single Linkage = Nearest Neighbour

$$d(C_1, C_2) = \min_{\mathbf{x}_1 \in C_1; \mathbf{x}_2 \in C_2} d(\mathbf{x}_1, \mathbf{x}_2),$$

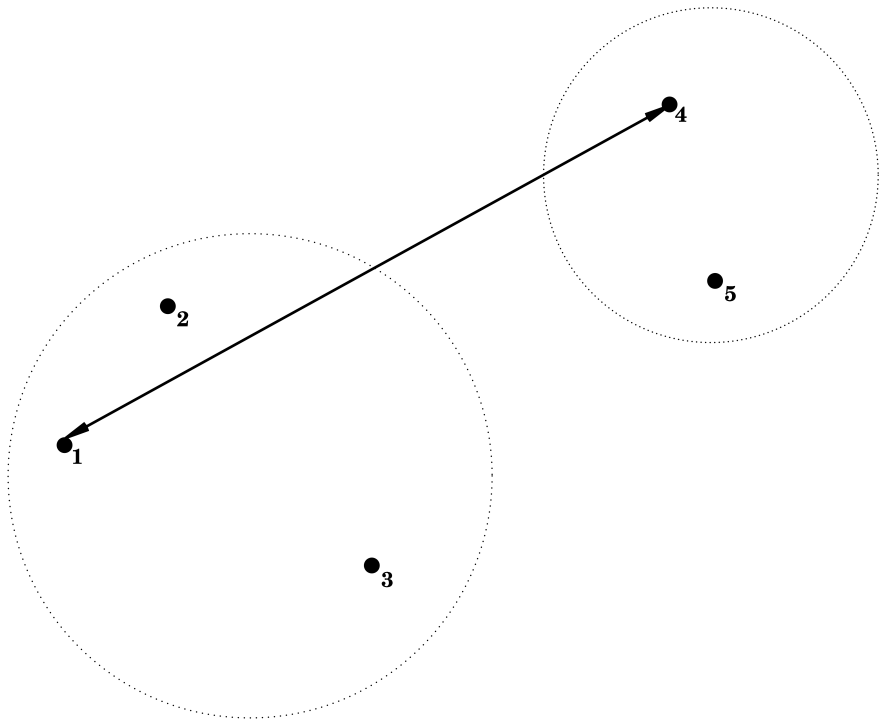
i.e. the dissimilarity between $C_1$ and $C_2$ is determined by the smallest dissimilarity between a point of $C_1$ and a point of $C_2$.



### 3.2.2 Complete Linkage = Furthest Neighbour

$$d(C_1, C_2) = \max_{\mathbf{x}_1 \in C_1; \mathbf{x}_2 \in C_2} d(\mathbf{x}_1, \mathbf{x}_2),$$

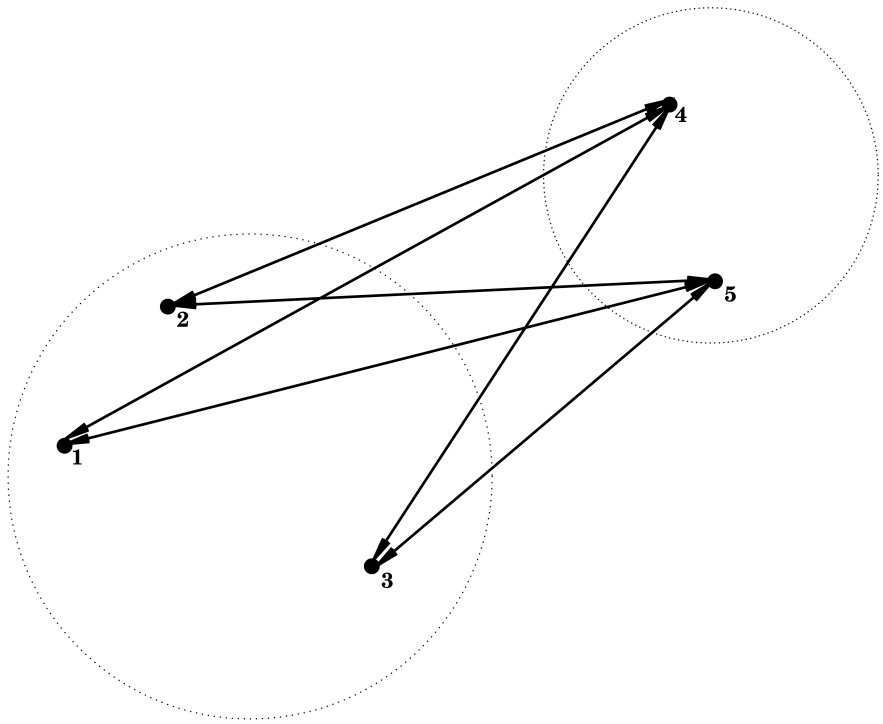
i.e. the dissimilarity between $C_1$ and $C_2$ is determined by the largest dissimilarity between a point of $C_1$ and a point of $C_2$.




### 3.2.3 Average Linkage = Group Average

$$d(C_1, C_2) = \frac{1}{|C_1||C_2|} \sum_{\mathbf{x}_1 \in C_1; \mathbf{x}_2 \in C_2} d(\mathbf{x}_1, \mathbf{x}_2),$$

i.e. the dissimilarity between $C_1$ and $C_2$ is determined by the average dissimilarity between all points of $C_1$ and all points of $C_2$.




### 3.3 Cluster Tree

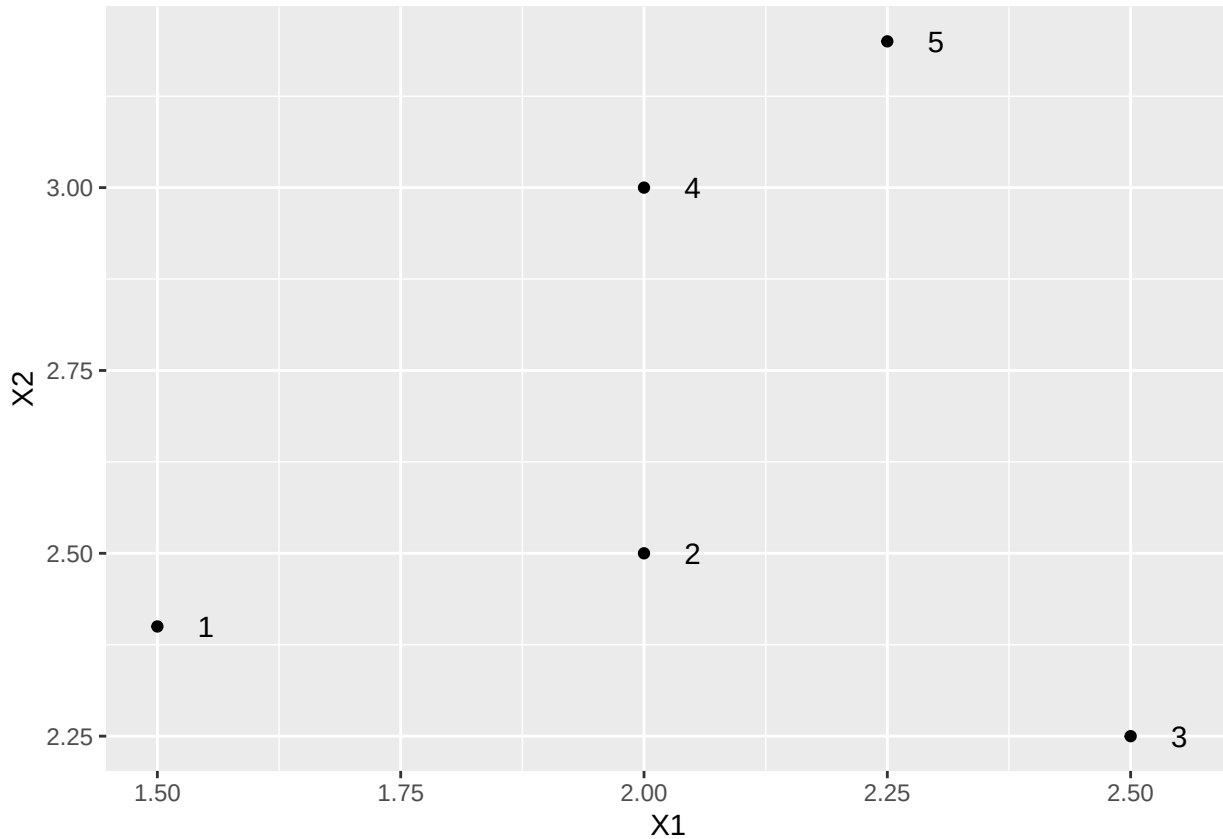
Hierarchical nature of the algorithm:

- Nested sequence of clusters $\longrightarrow$ visualisation via a tree
- Height of branches indicate the intercluster dissimilarity at which clusters are merged.
- Can used as instrument for deciding the number of clusters in the data

## 4 Toy example

| X1 | X2 | label |
|---|---|---|
| 1.50 | 2.40 | 1 |
| 2.00 | 2.50 | 2 |
| 2.50 | 2.25 | 3 |
| 2.00 | 3.00 | 4 |
| 2.25 | 3.20 | 5 |

```
toy %>%
  ggplot(aes(X1, X2, label = label)) +
  geom_point() +
  geom_text(nudge_x = .05)
```

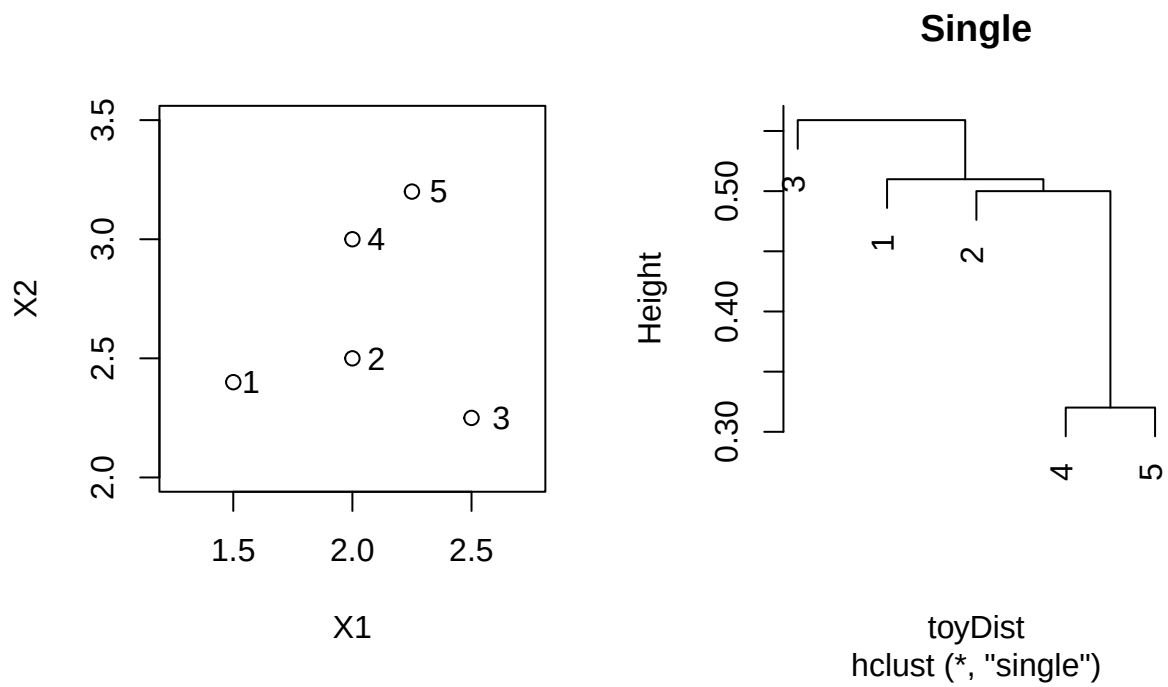



```
toy[,1:2] %>% dist
```

```
##           1         2         3         4
## 2 0.5099020
## 3 1.0111874 0.5590170
## 4 0.7810250 0.5000000 0.9013878
## 5 1.0965856 0.7433034 0.9823441 0.3201562
```

### 4.1 Single linkage

```
toyDist <- toy[,1:2] %>% dist
toySingle <- hclust(toyDist, method = "single")
par(mfrow=c(1,2),pty="s")
plot(X2 ~ X1, toy, xlim = c(1.25,2.75),ylim = c(2,3.5))
text(toy$X1*1.05,toy$X2,label=toy$label)
plot(toySingle, main = "Single")
```

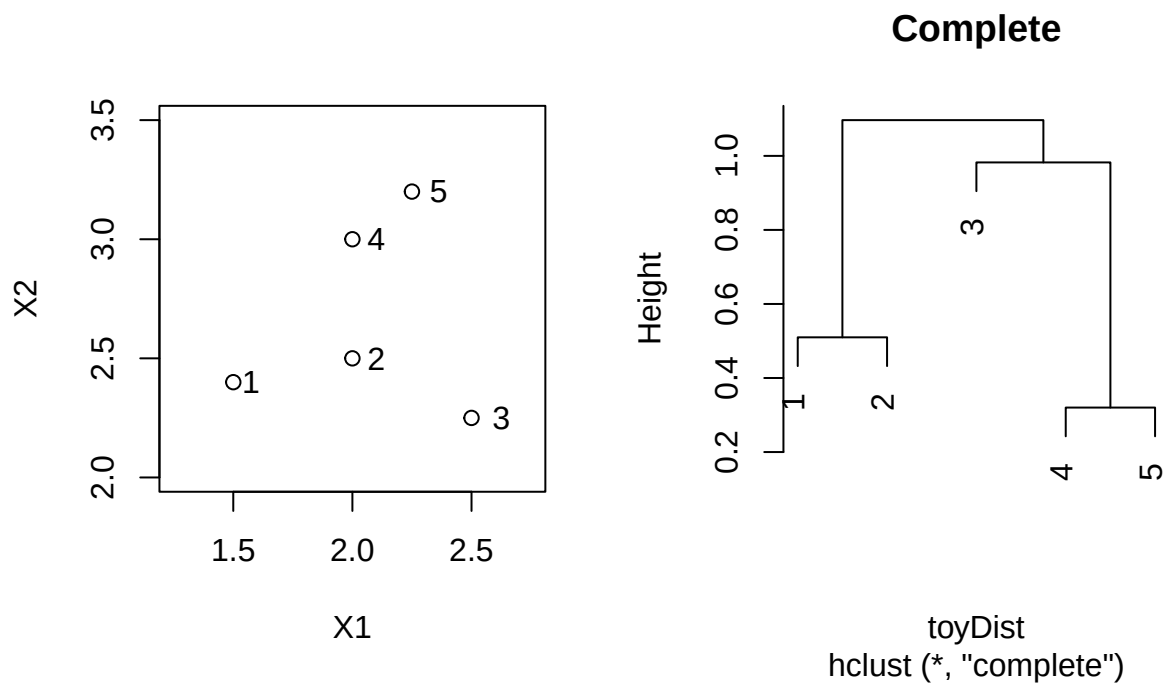



```
toyDist
```

```
##           1         2         3         4
## 2 0.5099020
## 3 1.0111874 0.5590170
## 4 0.7810250 0.5000000 0.9013878
## 5 1.0965856 0.7433034 0.9823441 0.3201562
```

### 4.2 Complete linkage

```
toyComplete <- hclust(toyDist, method = "complete")
par(mfrow=c(1,2),pty="s")
plot(X2 ~ X1, toy, xlim = c(1.25,2.75),ylim = c(2,3.5))
text(toy$X1*1.05,toy$X2,label=toy$label)
plot(toyComplete,  main = "Complete")
```

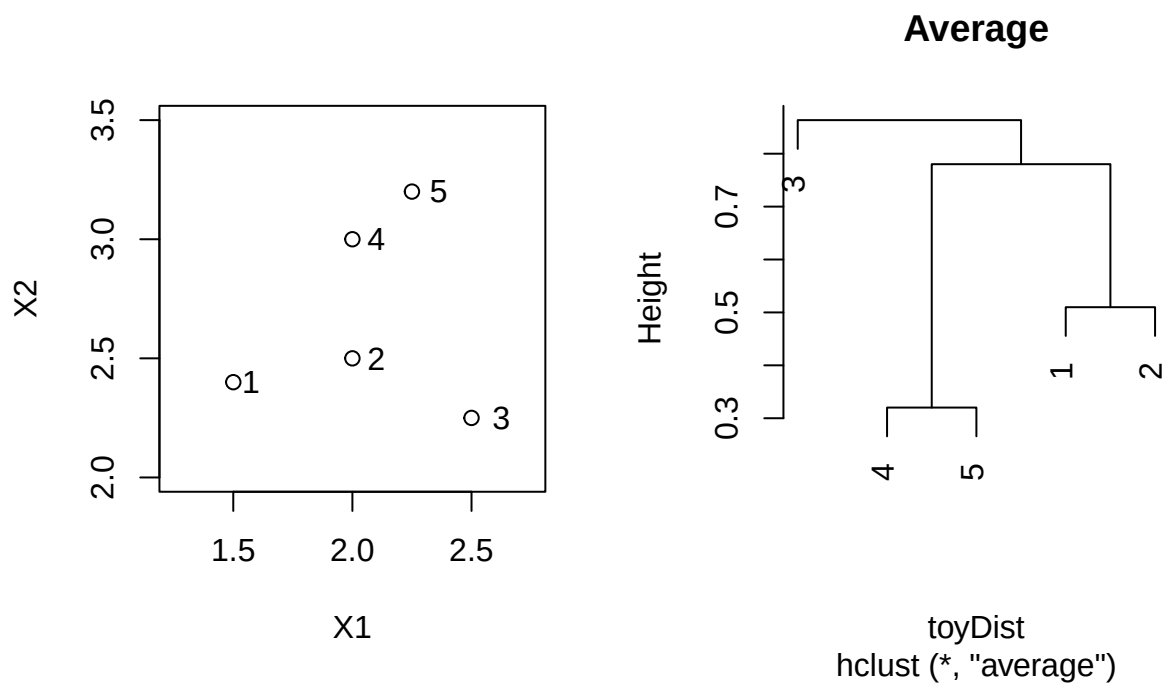



```
toyDist
```

```
##           1         2         3         4
## 2 0.5099020
## 3 1.0111874 0.5590170
## 4 0.7810250 0.5000000 0.9013878
## 5 1.0965856 0.7433034 0.9823441 0.3201562
```

### 4.3 Average linkage

```
toyAvg <- hclust(toyDist, method = "average")
par(mfrow=c(1,2),pty="s")
plot(X2 ~ X1, toy, xlim = c(1.25,2.75),ylim = c(2,3.5))
text(toy$X1*1.05,toy$X2,label=toy$label)
plot(toyAvg, main = "Average")
```

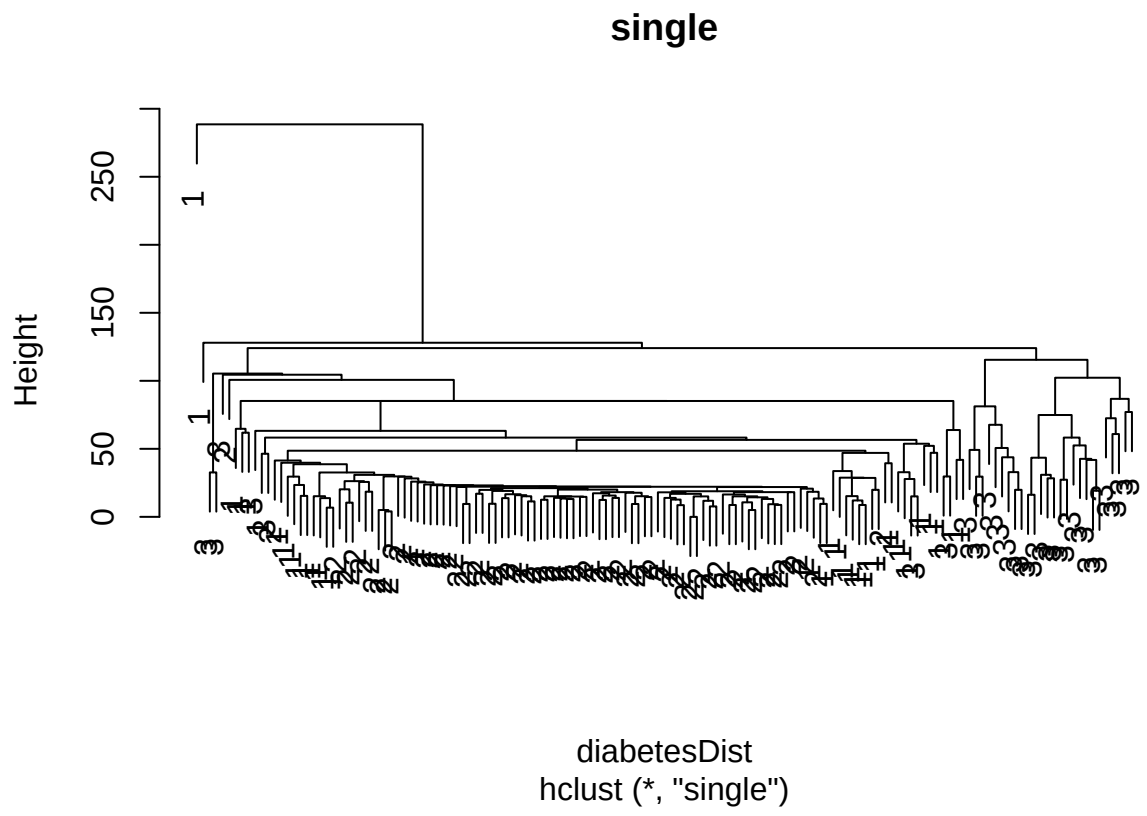



```
toyDist
```

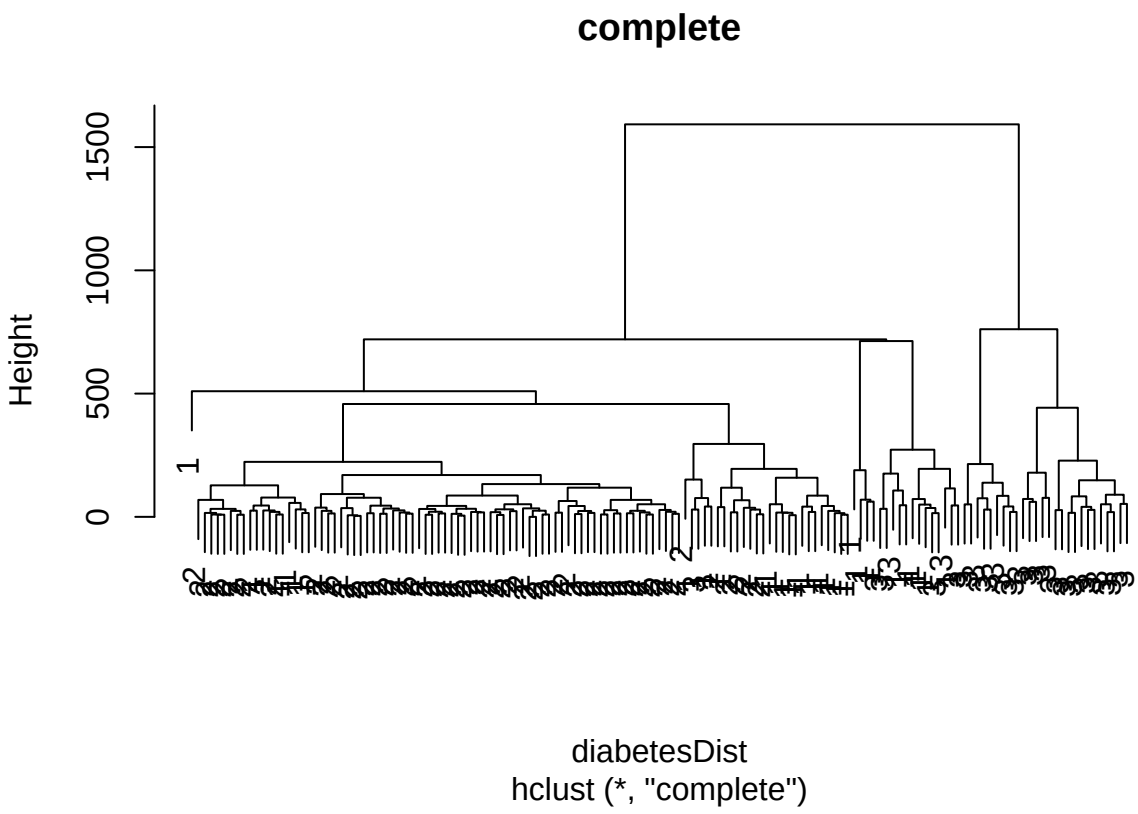
```
##           1         2         3         4
## 2 0.5099020
## 3 1.0111874 0.5590170
## 4 0.7810250 0.5000000 0.9013878
## 5 1.0965856 0.7433034 0.9823441 0.3201562
```

### 4.4 Example

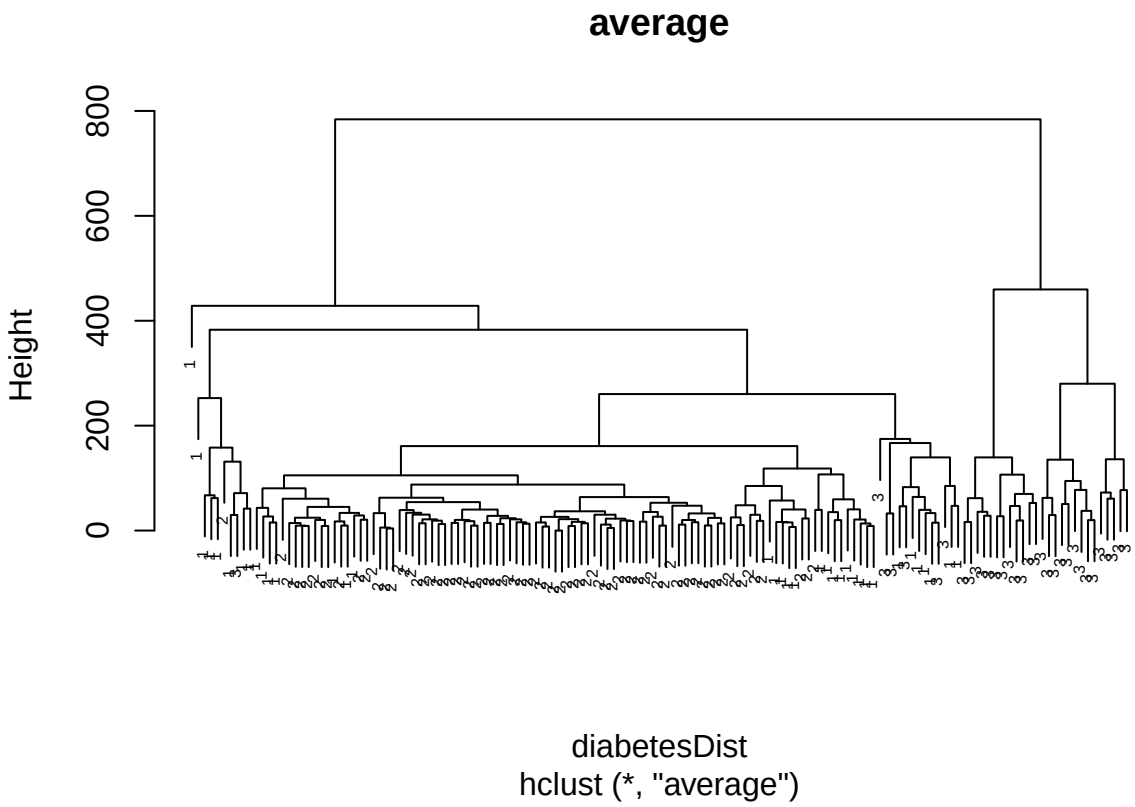
```
diabetesDist <- dist(diabetes[,-1])
diabetesSingle <- hclust(diabetesDist, method = "single")
plot(diabetesSingle, labels = as.double(diabetes$class), main="single")
```




```
diabetesComplete <- hclust(diabetesDist, method = "complete")
plot(diabetesComplete, labels = as.double(diabetes$class), main="complete")
```




```
diabetesAverage <- hclust(diabetesDist, method = "average")
plot(diabetesAverage, labels = as.double(diabetes$class), main = "average",cex=0.5)
```




# 5 Model-based clustering

- Paper: Fraley and Raftery (1998). How Many Clusters? Which Clustering Method? Answers Via Model-Based Cluster Analysis. The Computer Journal, (41)8:578-588.
- EM algorithm [PDF]
- Example: see tutorial session